Interactive Lenses for Visualization

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Introduction

What is an Interactive Lens

- Solve a localized visualization problem
- Temporarily altering a selected part of a visualization

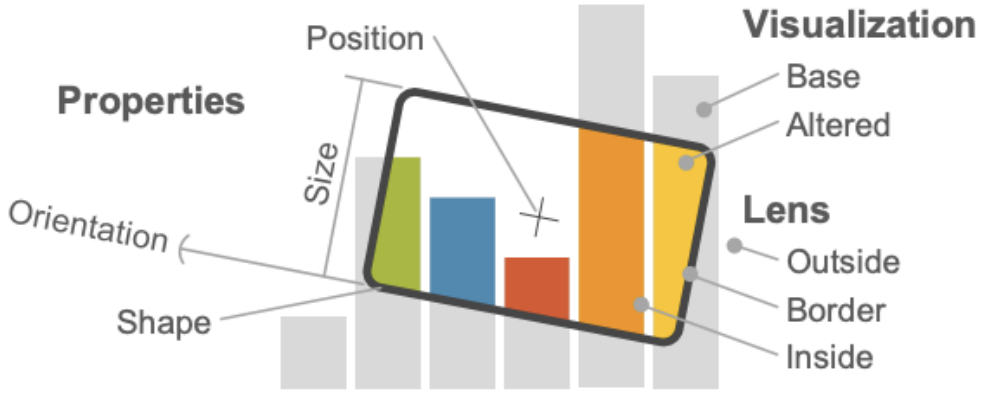
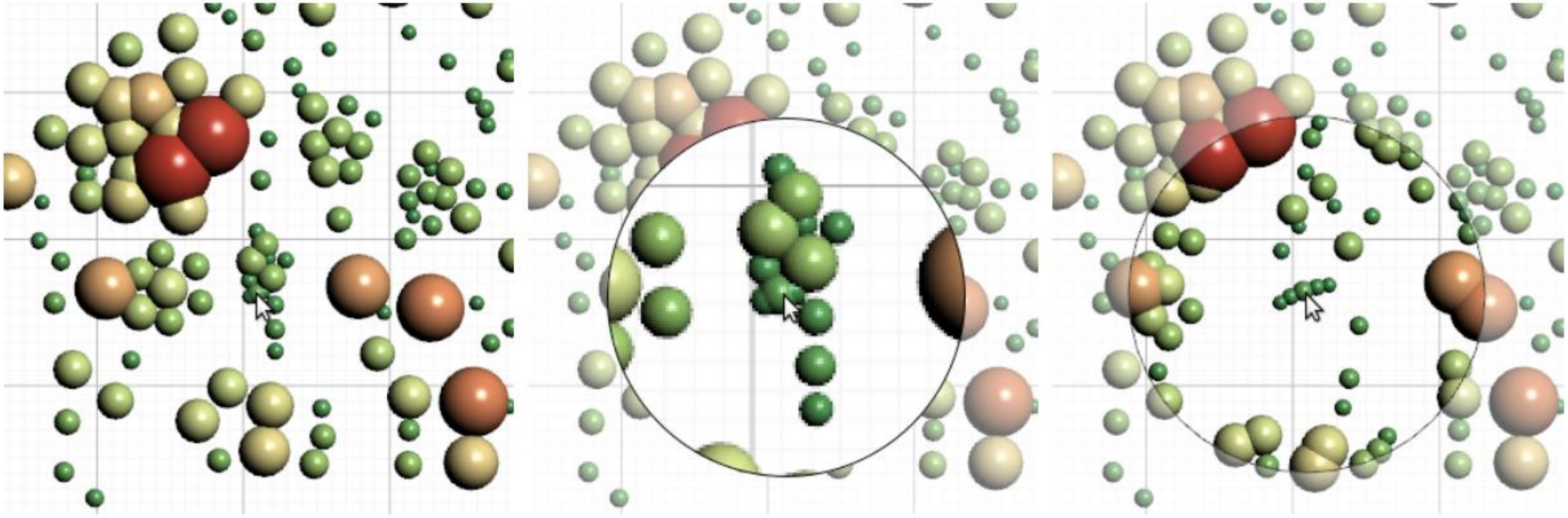


Figure 3: *Schematic depiction of an interactive lens.*

Why use an Interactive Lens

- Support method for data exploration
- Cluttered visualization
- Overview+Detail
- Focus+Context

Magnification Lens Example



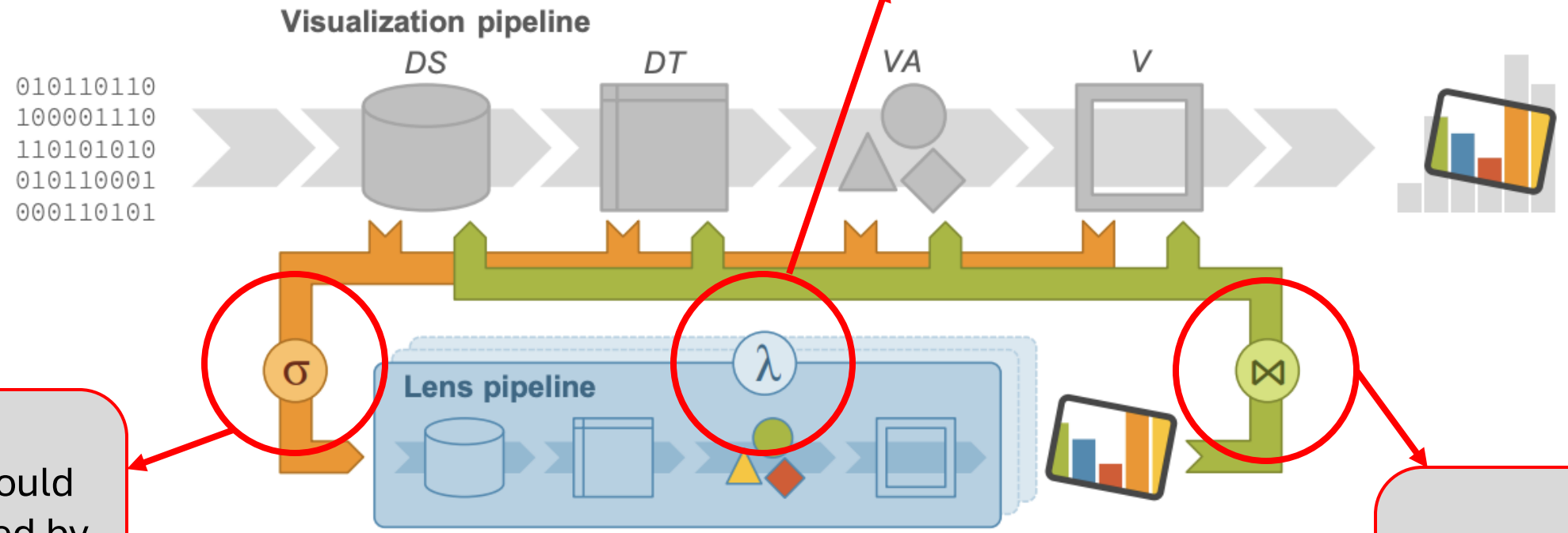
(a) Regular visualization

(b) Simple magnification

(c) Fish-eye distortion

Conceptual Model

How the visualization is modified

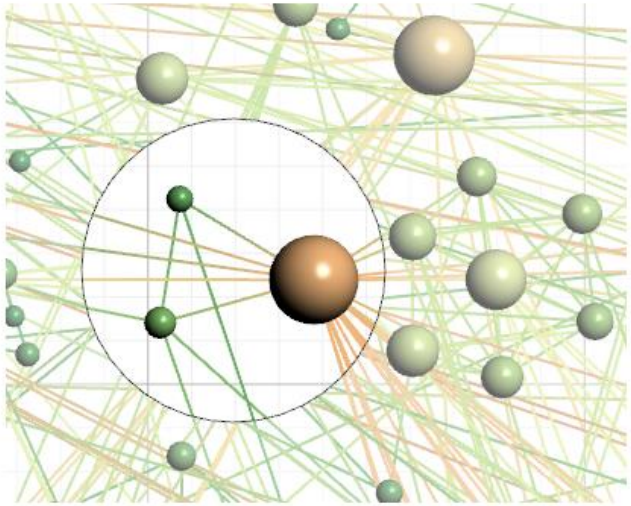


What should be affected by a Lens

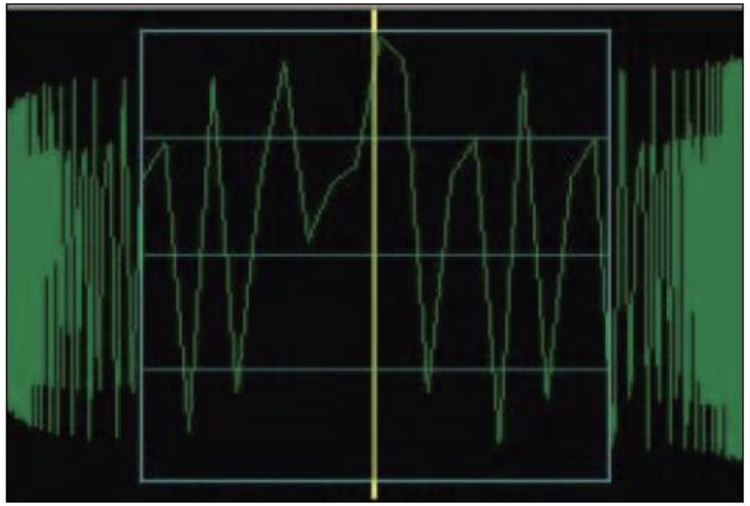
Result to be joined with base

Lens Properties

Shape



(a) Circular shape



(b) Rectangular shape

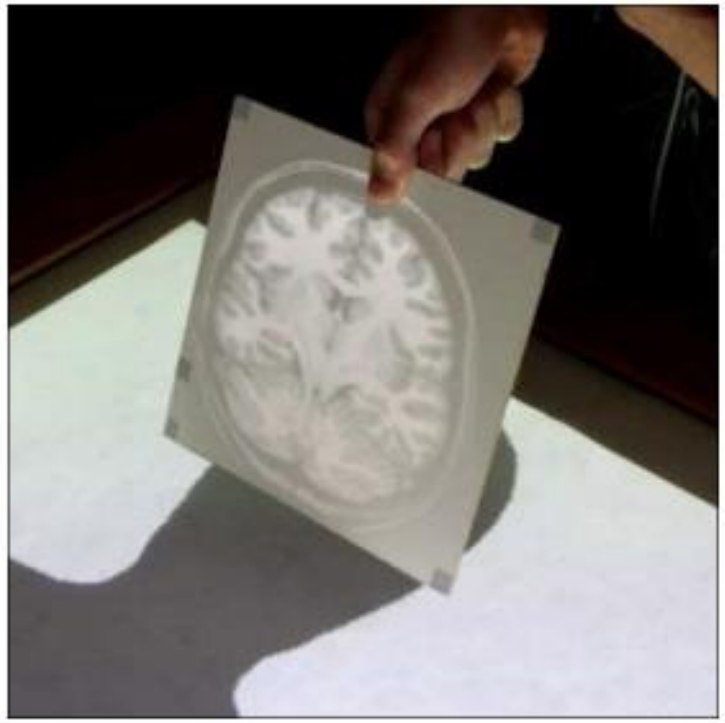


(c) Content-adaptive shape

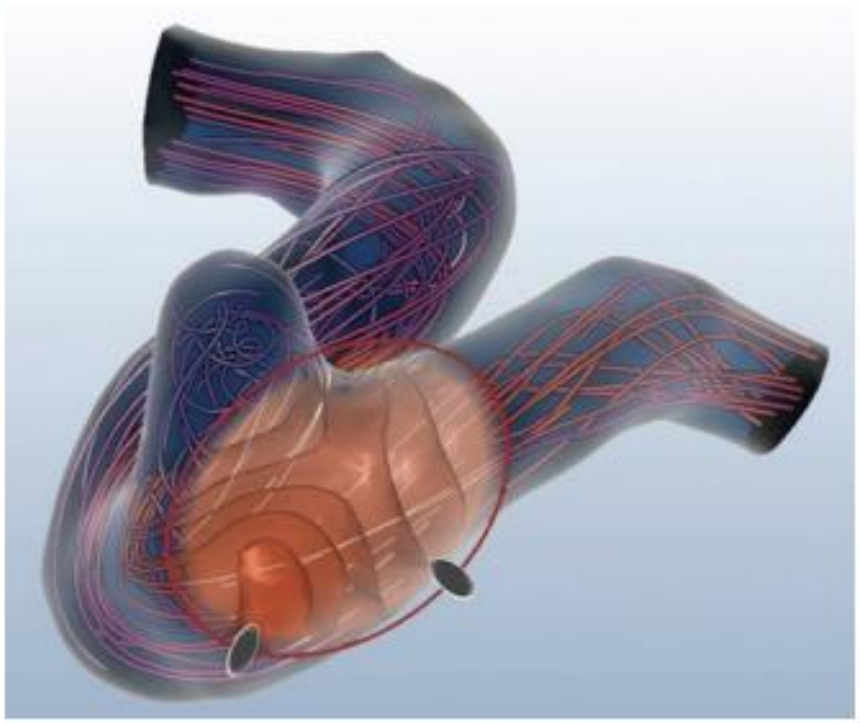
Position and Size

- Set or adjustable size
- Automatic sizing
- Mouse controlled position
- Area selection

Orientation



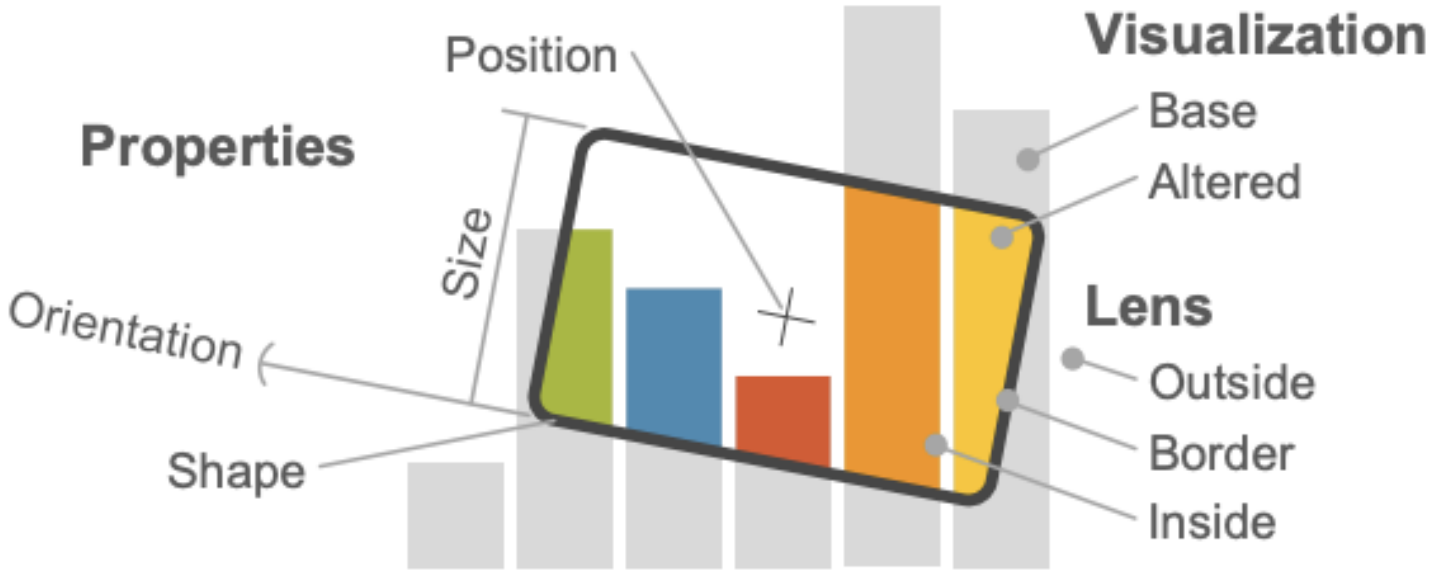
(a) *PaperLens* [SSD09]



(b) *FlowLens* [GNBP11]

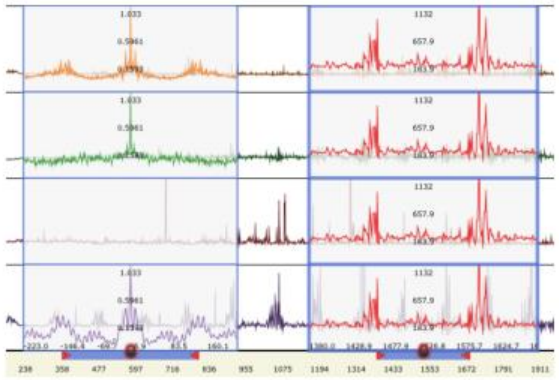
Lens Properties

Other



Data and Tasks

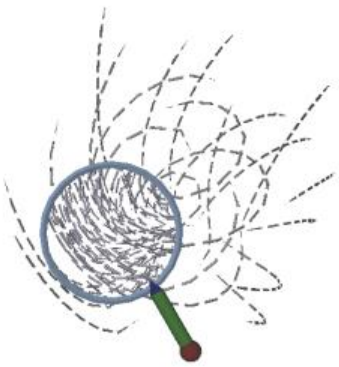
For Specific Data



(a) Temporal data



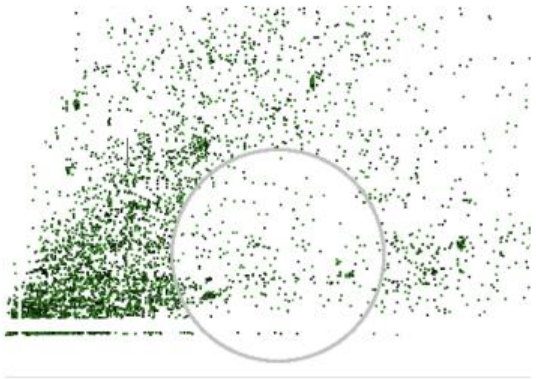
(b) Geospatial data



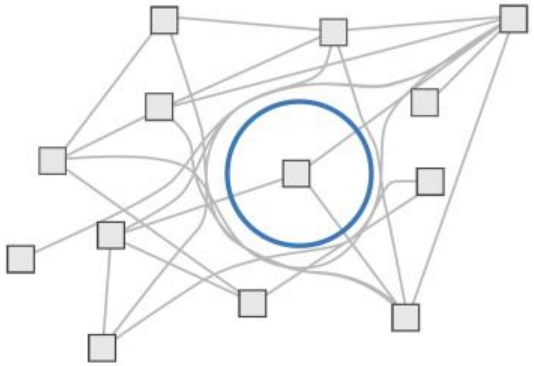
(c) Flow data



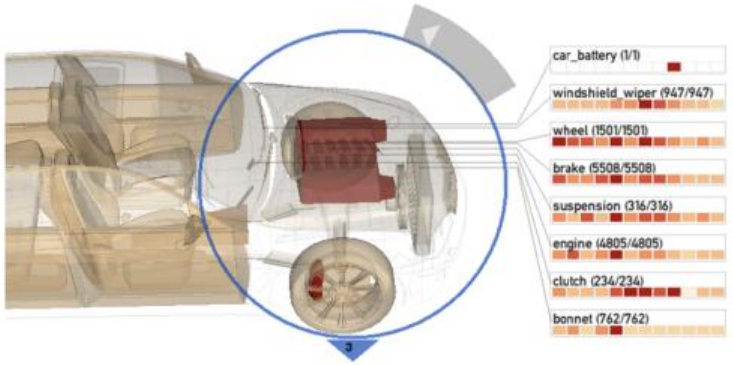
(d) Volume data



(e) Multivariate data

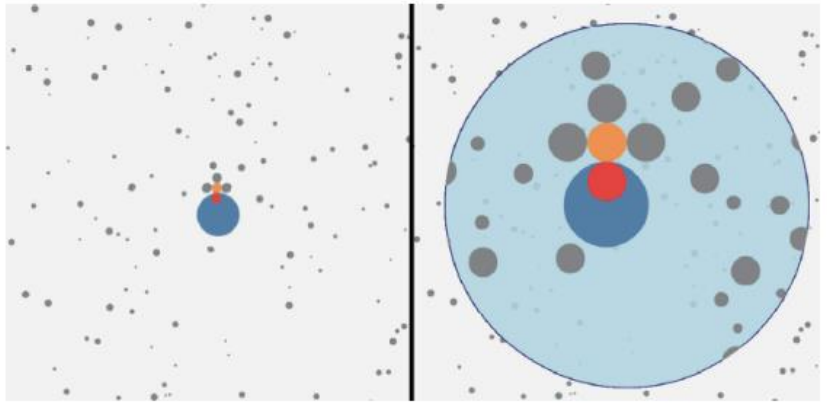


(f) Graph data

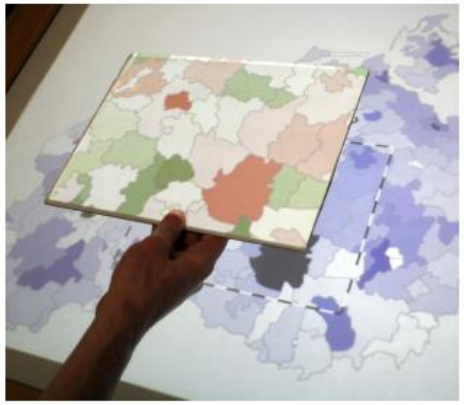


(g) Text and document data

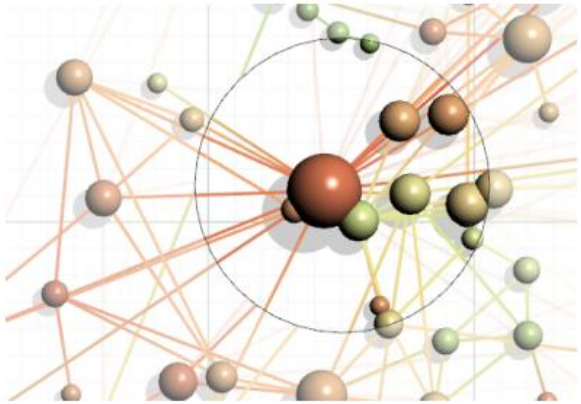
For Specific Tasks



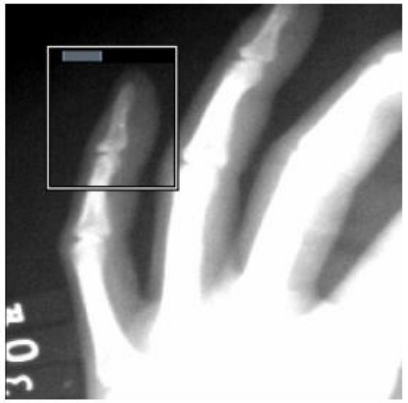
(a) Select



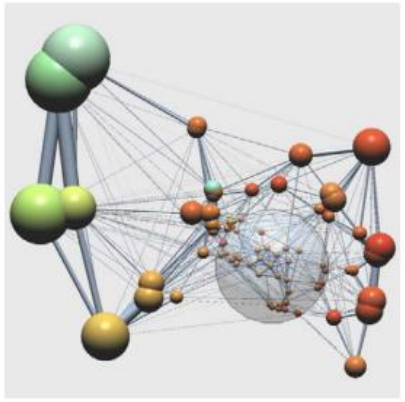
(b) Explore



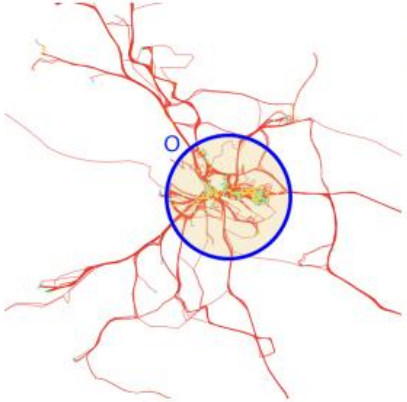
(c) Reconfigure



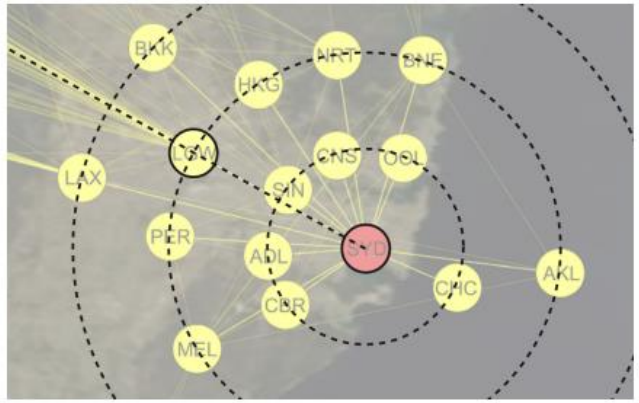
(d) Encode



(e) Abstract/Elaborate



(f) Filter

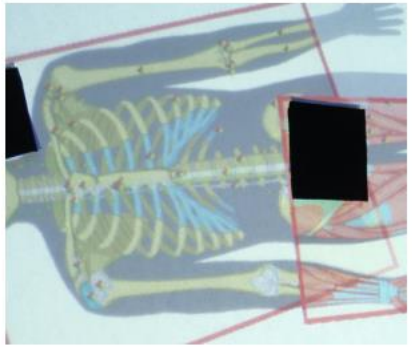


(g) Connect

Interaction

Interaction and Display

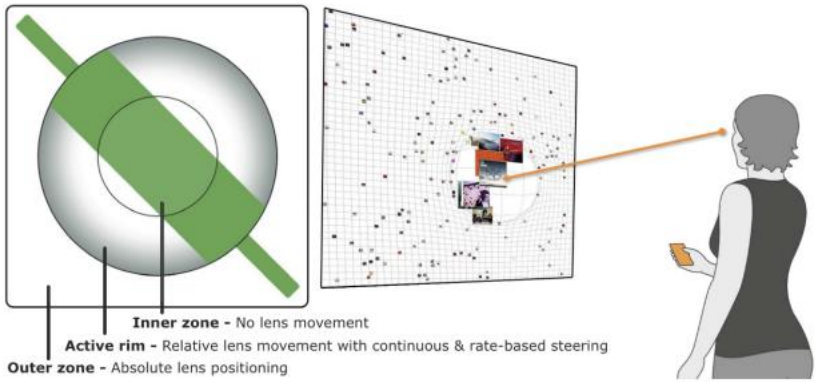
- Mouse and keyboard
- Touch
- Tangible
- Tangible views and spatial interaction
- Gaze-based interaction and head tracking



(a) Lens transparencies

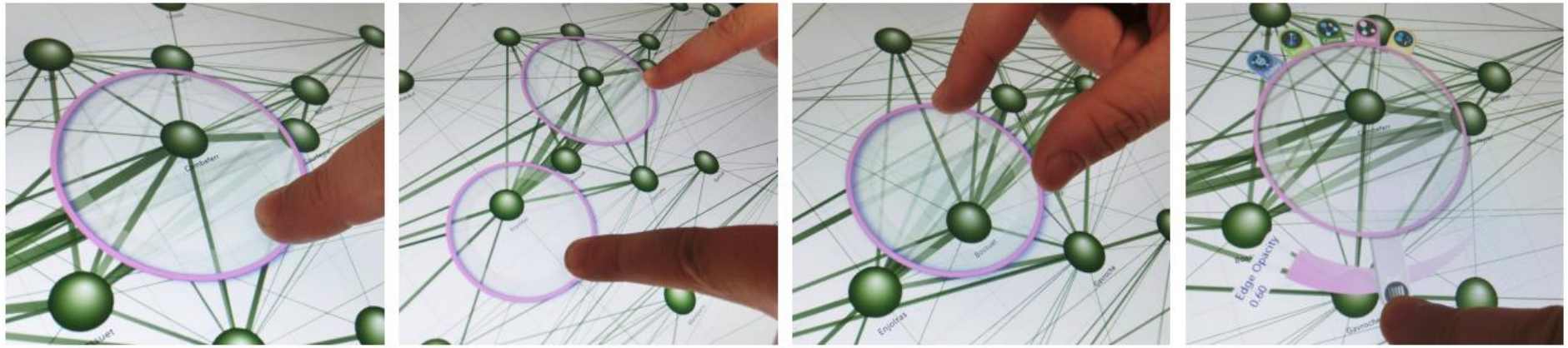


(b) Paper lenses

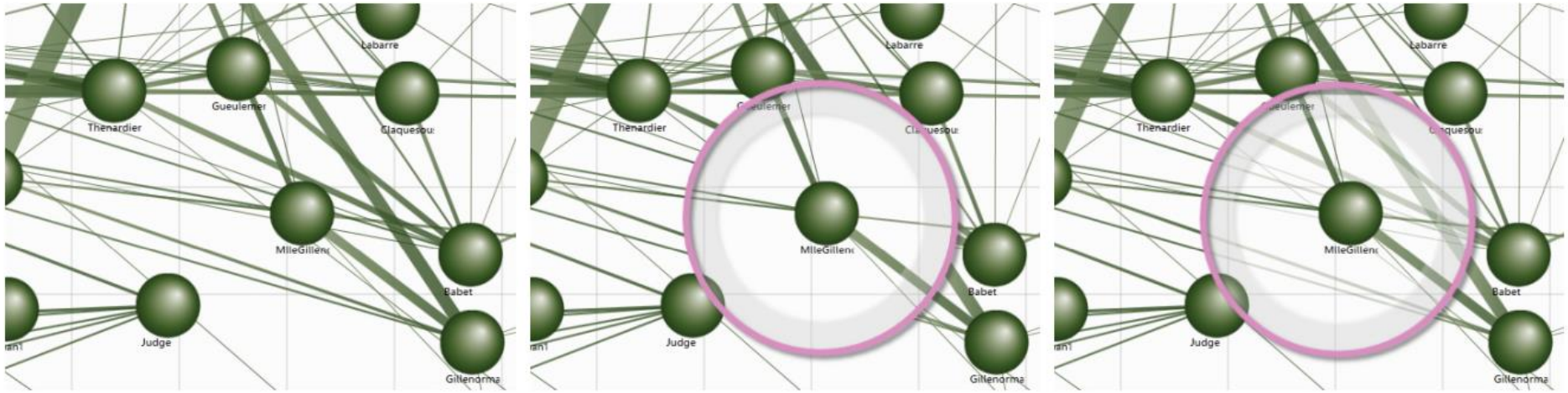


Example

Example



(a) Placing a single lens (b) Positioning multiple lenses (c) Resizing a lens (d) Adjusting lens parameters



(a) Cluttered node-link diagram (b) Suppressing irrelevant edges (c) Dimming irrelevant edges

Future Of Interactive Lenses

Future Directions

- Interaction improvements
- Collaborative multi-user work
- Lenses for data manipulation
- *Implement-once-and-reuse-many-times*